Introduction to COMP 206
Lab A

The labs for this course are designed to be completed either independently at home or in the Trottier 3rd-floor labs. These labs are not graded, and you are not required to submit them. If you would like to work on a lab with the help of TA, please check the lab schedule posted in the #schedule channel on Discord or on Minerva to find available lab time slots. Since university lab space is limited, your TA may ask you to bring your own laptop instead of using a lab computer. This lab is about getting used to the School of Computer Science servers and the Linux command line.

Some labs will have a question zero. These questions will not be covered by the TA during the tutorial. It is extra content meant for you to do on your own.

At the end of this lab is a Mentoring section. The TA will use this during their tutorial section.

QUESTION ZERO: Optional problem

(A) Let us learn a little about the Linux Manifesto.

Google "Linux Documentation Project Manifesto". Summarize the manifesto in two to four sentences.

(B) Popularity of development environments

There are two common environments that run software: the server and the personal device. Developers need to know where their software will be deployed in order to properly construct their algorithm so that it is sensitive to the deployed environment. Servers have different requirements compared to personal computers/phones. In this course we will develop software for both personal computers and servers.

Look at the following two links and determine the popularity of Windows vs Apple vs Linux for (A) used as a personal computer and (B) used as a server.

The two links are: https://www.w3schools.com/browsers/browsers_os.asp and https://en.wikipedia.org/wiki/Usage_share_of_operating_systems

QUESTION ONE: Getting and using your CS user account

It is important that you establish the following procedure when submitting assignments:

1. Use the **mimi.cs.mcgill.ca** server for all assignments. Only this server will be accepted for assignments. The TAs access this server for grading. You can **ssh** into this machine or go to Trottier 3rd floor to work on this machine, to create your assignments. All computers on the third floor of Trottier can be used. You can also remotely connect using iOS **ssh** (at the command line of your PC) or using Windows **cmd** or **putty** (download the app for your PC).
    a. HOWEVER! You **cannot log** into this server using your first.last@mail.mcgill.ca account because that will redirect you to the **McGill** server and not to **mimi.cs.mcgill.ca** even though you logged in using a computer in Trottier or logged in from home by typing `ssh mimi.cs.mcgill.ca !!`
    b. You must first create a computer science user account by doing the following:
        i. Make sure you are on campus (or using a VPN)
        ii. Open your browser and type: https://www.cs.mcgill.ca/docs/ and follow the instructions under CS Accounts and Account Creation.
2. Now try to login.
    a. If you are on your own **Mac or Linux** device, go to the command line (For Mac: launchpad then **Terminal** or search Terminal, for Linux: Ctrl+Alt+T or search Terminal) and type `ssh username@mimi.cs.mcgill.ca` , you should see a prompt asking you for your password. If all goes well, you will then see the welcome message and a prompt from the server. Note that if you only type `ssh mimi.cs.mcgill.ca`, your terminal will most likely assume that the username is the same as your computer's name, which is probably wrong. Depending on your terminal, you might be prompted for a username.
    b. If you are in the Trottier lab, you will see a GUI login screen. Enter your CS user name and password. If all goes well, you will see the desktop. You can then find the terminal and `ssh` as described above for Linux.
    c. If you are home with a **Windows** computer, you can open your terminal like Mac/Linux users but it is called **cmd**. From that terminal you can use the **ssh** command. Alternatively, if **ssh** is not installed by default in your **cmd**, then download the open source program named **Putty** on https://www.putty.org/. Then use the GUI to input your username, password, and the URL **mimi.cs.mcgill.ca**. Then press the login or connect button. If all goes well, you should see the command line and welcome message.

2.1 Passwordless Login to Mimi (Useful for Remote Access)

You can log in to Mimi without entering your password each time by setting up an SSH key. An SSH key is a special file consisting of two parts: a public key and a private key. The public key can be shared with others, while the private key must always be kept secret. Together, they enable secure authentication as follows:

1. You install your public key on your Mimi account.
2. When you try to log in, the server uses your public key to create a "challenge" and sends it to your computer.
3. SSH uses your private key to solve the challenge and sends back proof.
4. The server verifies the proof (which only your private key could produce) and grants you access.

On Mimi, username-and-password authentication works only when you are on McGill Wi-Fi or connected to the McGill VPN. However, key-based authentication works under any circumstances.

SSH Key Setup

To set up key-based authentication:

- On your computer, run ssh-keygen and follow the prompts to generate a key pair.
    - The keys are saved in ~/.ssh/id_rsa (private key) and ~/.ssh/id_rsa.pub (public key).
    - Important: Never share or upload your private key (id_rsa). Only the public key (id_rsa.pub) should be shared.
- Next, run ssh-copy-id username@mimi.cs.mcgill.ca to install your public key on Mimi. (You must be on the McGill network for this step.)
- If successful, you should now be able to log in with:
- ssh username@mimi.cs.mcgill.ca without being prompted for a password.

3. Now, let us logout
    a. If you are using a Mac, then type `logout` and your connection with the server ends. Then click on the X to close the Terminal window.
    b. If you are using Linux (which includes the Trottier lab) and in the command line window then type `exit` to close the window or press the X to close the window. Then you will need to logout of the desktop. On the right top of the screen (usually) is the logout menu option. Select it.
    c. If you are at home using Windows cmd, then type `logout` and your connection with the server ends. Then click on the X to close the cmd window. If you used Putty, then press X to close the Putty program.

QUESTION TWO: Backing up Files

Try this experiment. If you want more information on a command, you can use `man` followed by the command name

(A) Login to **mimi.cs.mcgill.ca**
(B) In your home directory create an empty file called **test.txt** by using `touch`
(C) `mkdir` a directory called **LabA**
(D) Then `cd` into that directory and `mkdir` a subdirectory called **backup**
(E) Using `cp` copy a random file (can be test.txt) from your home directory into LabA renaming it to **letter1.txt** Do not worry that you are changing the file's name within the LabA subdirectory. It will not affect the original file.
(F) Using `cp` to copy another random file into LabA (or test.txt again), renaming it to **letter2.txt**.
(G) From within the LabA subdirectory, use `touch` to create a third file called **letter3.txt** in LabA.
(H) Look at the files in the LabA directory by typing `ls`. You can also try `cat filename` to see the

content of a specific file: In this case, your files might be empty if these are copies of test.txt

(I) Using TAR archive your three text files into the directory backup. You will need to write something like this: `tar -cvf backup/myletters.tar *.txt` This command takes all your text files (*.txt) and merges them into an archive file called myletters.tar within the subdirectory backup.

(J) Using `ls`, look at the directory backup to see if your myletters.tar file exists. If it does, then you did it correctly.

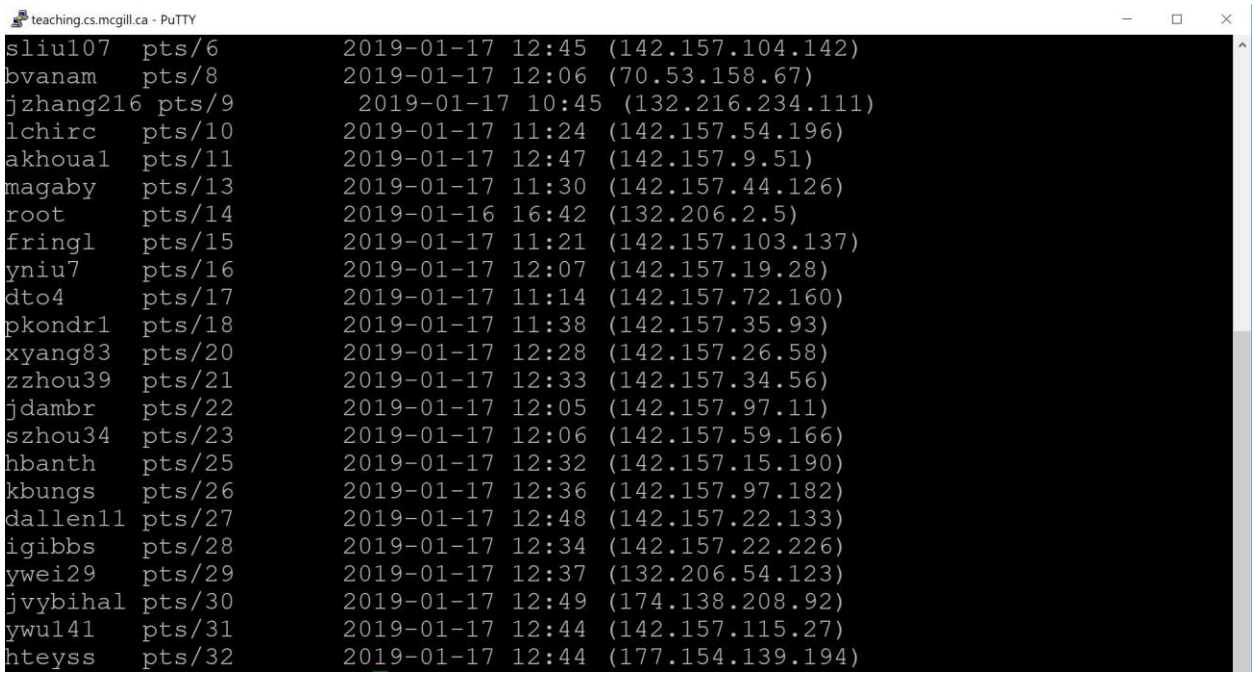
(K) Now, `cd` into the archive directory and extract the files by typing `tar -xf myletters.tar`. The files you archived should be extracted within the same directory (the **backup** directory). Once extracted they should appear within the directory.

(L) If something unexpected occurred while archiving / extracting, discuss this with the TA.

P.S. We will cover these commands shortly in class, but it's good practice to explore the man pages to understand their functionality and available options.

QUESTION THREE: Viewing who else is on the server

You can Find out the users currently logged in by typing the command: `who` or `w` for short. Notice that the command `who` displays something like this:



You can also view your username by typing the `whoami` command

You have completed the lab.

MENTORING SECTION

Begin the tutorial by reviewing some information:

- Connect your laptop to the project to give a demonstration to the students on how you open your laptop's terminal and log into the SOCS server using ssh. Let them see how you accessed the terminal, typed ssh, and logged out of mimi.
- Now do Question 1 from lab A. Help everyone get a SOCS account and complete Question 1.
- Give a demonstration on how to use FTP to transfer files from your laptop to the server and then from the server back to your laptop. You may need to write the command line syntax on the board. The FTP command line program should be installed on both Mac and Windows machines. You can also tell them about the FileZilla Client and the WinSCP client for windowed file transfers.

Now focus on the remaining lab questions:

- Ask the students whether they would like you to do all the questions or to just focus on specific questions. I am assuming you completed the above "reviewing some information" section. Then ask the class if they would like you to do it for them step-by-step or whether they should do it on their own and you watch/help.
- If the class would like you to do it step-by-step for them, then you will need to ask them to open their laptops. Connect your laptop to the projector. You will do each step of the lab pausing at the end of each step for the students to also complete that step. Try to keep the class synchronized. Answer questions and debug as you go along.
- If the students want to do it on their own, then walk around and be available to help them.

If there is not enough time to complete the lab, the students will need to complete the lab on their own. But the students can come during office hours for help.